

# ADRIAN SALAVATY

Melbourne, Australia

[www.ASalavaty.com](http://www.ASalavaty.com)

[github.com/asalavaty](https://github.com/asalavaty)

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## SKILLS

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- **Bioinformatics:** High-throughput data analysis including transcriptomics (bulk, sc, spatial RNA-seq and Microarray), genomics (WGS/WES), and epigenomics (Chip-seq); Proteomic (Mass Spec.) data analysis; Multi-omics (IntegratOMIC) data analysis, Nextflow pipeline development
- **Systems Biology:** Pathway and gene-set data analysis; Single-layer (*e.g.*, co-expression, PPI, genomic co-deleteriousness, and metabolomic co-abundance) and multi-layer (multi-omics) network reconstruction, analysis, and visualization; Node centrality and influence evaluation, Graph-based model development; Node centrality algorithm development and optimization
- **Immunoinformatics:** Drug/compound dose-response analysis, FACS cell count and marker expression data analysis and visualization, IncuCyte GOC/ROC, CTG, RII, and AlphaLISA data analysis and visualization, High-throughput immune-response compound screening data analysis
- **Cheminformatics:** Drug/compound molecular fingerprint analysis, structural similarity analysis, drug-drug interaction prediction, drug-target interaction analysis
- **Programming languages and skills:** R (tidy R programming), R package development, Shiny app development, Python, Python package development, Nextflow, Linux and command-line-based high performance computing (HPC)
- **AI and Machine learning (ML):** AI Agent Development, Supervised and unsupervised ML-based problem solving such as clustering, classification, regression, and dimension reduction; Experienced in classical (*e.g.*, SVM and Random Forest) and modern (Neural Network-based) ML models; Feature selection/extraction
- **Statistics:** Biostatistics, Data mining, Meta-analysis
- **Data visualization:** R-based (ggplot2, etc.) and Python-based (matplotlib, seaborn, plotnine, etc.) advanced data visualization
- **Scientific writing:** Reference manager (EndNote, Mendeley, Citavi)
- **Computer skills:** MS Office, Adobe Photoshop, Adobe Illustrator

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## CERTIFICATES

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- Data Science and Machine Learning: Making Data-Driven Decisions  
MIT Institute for Data, Systems, and Society (IDSS)

Sep 2023

- Python Essential Training  
LinkedIn Learning Feb 2022
- Learning Python  
LinkedIn Learning May 2021
- Application of NGS in Cancer Diagnosis and Management  
Isfahan University of Medical Sciences, Department of Genetics and Molecular Biology  
OMICSTM October 2015  
BGI Tech Isfahan, Iran
- Bioinformatics Workshop  
Shahid Chamran University of Ahvaz June 2014  
Ahvaz, Iran

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## PROGRAMMING PROJECTS

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### **Shiny Apps**

- AutoClone: Calculation of distances (clonality) based on color features  
URL: <https://autoclone.erc.monash.edu/>  
Monash University 2021
- Influential Software Package web portal  
URL: <https://influential.erc.monash.edu/>  
ExIR Shiny app  
IVI Shiny app  
Monash University 2021

### **Python-based Packages and Models**

- influential: Identification and Classification of the Most Influential Nodes 2023  
Python package, released on PyPI  
PyPI link: <https://pypi.org/project/influential/>  
GitHub repo: <https://github.com/asalavaty/python-influential>  
Experimental-data-based Integrative Ranking (ExIR)  
Integrated Value of Influence (IVI)  
SIR model-based Influence Ranking (SIRIR)

### **R-based Packages and Models**

- CelliVerse: An Ecosystem for Exploring the Universe of Single-Cell Data  
R package, released on CRAN  
CRAN link: <https://cran.r-project.org/package=celliverse>  
Website: <https://asalavaty.github.io/celliverse/>  
GitHub repo: <https://github.com/asalavaty/celliverse>

ClustoCell, MarkoClust, MarkoCell, MarkerPurity

Peter Mac Cancer Center

2025-2026

- influential: Identification and Classification of the Most Influential Nodes

R package, released on CRAN

CRAN link: <https://cran.r-project.org/package=influential>

Website: <https://asalavaty.github.io/influential/>

GitHub repo: <https://github.com/asalavaty/influential>

Experimental-data-based Integrative Ranking (ExIR)

Integrated Value of Influence (IVI)

SIR model-based Influence Ranking (SIRIR)

Monash University

2020-2021

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## EDUCATION

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### Ph.D. in Bioinformatics

2019-2022

Australian Regenerative Medicine Institute, Monash University

Melb, AUS

PhD thesis: Identification of commonalities in clonal selection during normal and cancer tissue development using bioinformatics and systems biology techniques

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*Skills: Bioinformatics (bulk and scRNA-Seq), Systems Biology (network analysis and network-based model development), R and Python Programming, Shiny App, Machine Learning, HPC*

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### Master of Science in Biology-Biochemistry

2016-2018

University of Kashan

Kashan, Iran

Master thesis: Computational functional analysis and annotation of lung adenocarcinoma prognostic long non-coding RNAs

GPA: 3.82/4

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*Skills: Bioinformatics (microarray and RNA-Seq), Systems Biology (network analysis, pathway and gene-set data analysis), R Programming, HPC, Data Visualization*

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### Bachelor of Science in Genetics

2011-2015

Shahid Chamran University of Ahvaz

Ahvaz, Iran

GPA: 3.72/4

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*Skills: Bioinformatics (microarray, meta-analysis, genomics), Systems Biology (pathway and gene-set data analysis), R Programming, Data Visualization*

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## PROFESSIONAL EXPERIENCES

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### Affiliations

- Senior Bioinformatician 2024-present  
Bioinformatics Core Facility, Peter Mac Cancer Centre, Melbourne, Australia

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*Skills: Bioinformatics (Bulk and scRNA-Seq, Spatial transcriptomics/proteomics, Multi-Omics Data Analysis), Systems Biology (network analysis and graph-based model dev.), Machine Learning, Shiny app dev.*

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- Senior Bioinformatic Scientist 2023-2024  
oNKO-innate, Melbourne, Australia

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*Skills: Bioinformatics (scRNA-Seq), Systems Biology (network analysis and graph-based model dev.), Drug-target compound screening analysis, Machine Learning, Nextflow pipeline dev., Shiny app dev., Text-mining*

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- Scientific Advisor, Arta Bioanalytics 2023- 2024
- Post-doctoral Research Officer (Bioinformatics and Systems Biology Scientist)  
Children's Cancer Institute Australia 2022-2023

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*Skills: Bioinformatics (Genomics and RNA-Seq), Systems Biology (network analysis and network-based model development, protein-protein interactions), R Programming, Text-mining, Machine Learning*

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- Scientific Advisor, ANUNA AI (Previously MEDDA) 2021-2023
- The Systems Biology Institute Australia 2019-present
- Professional Member of Cancer Epigenetics Society (ID Number: 1537) 2018-present
- Research fellow (Bioinformatician), Al-Zahra Medical Genetics Laboratory, 2015-2018  
Isfahan, Isfahan, Iran

### **Editorial and Reviewer**

- Reviewer, Nature - Scientific Reports 2023
- Reviewer, BMC Bioinformatics 2023
- Reviewer, Frontiers in Genetics 2020
- Reviewer, Journal of Rare Diseases Research & Treatment 2017
- Reviewer, Molecular Neurobiology 2017
- Editor, AMOR: Advances in Modern Oncology Research 2015- present
- Reviewer, MOJPB: MedCrave Online Journal of Proteomics & Bioinformatics 2014-2016

### **Teaching Experiences**

- TA of medical biotechnology; genomics, proteomics and bioinformatics (BRM5012) 2021  
Monash University Melbourne, Australia
- TA of Introduction to Bioinformatics (BMS5021) 2021  
Monash University Melbourne, Australia
- TA of Genomics and its applications (GEN3040) 2020-2021  
Monash University Melbourne, Australia
- Cancer Systems Biology Workshop 2018  
Tehran University of Medical Sciences Tehran, Iran

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## AWARDS AND HONORS

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- Awardee of Postgraduate Publication Award 2022  
Monash University top 30 PhD students according to publication records/prospects
- Awardee of the Best Oral Talk Based on People's Choice  
COMBINE-ABACBS Student Symposium 2021
- Awardee of the Best Student Oral Talk  
Victorian Cancer Bioinformatics Symposium 2021
- Awardee of Australia's Global Talent Permanent Residency Visa  
Data Science sector 2020
- Ph.D. scholarship from Monash University, (awarded and taken up) 2018

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## GRANTS

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- AstraZeneca Research Grant March 2026  
*Multi-omics investigation of mCRPC rapid autopsy samples*  
Role: Associate Investigator (AI)  
Chief investigator: Shahneen Sandhu
- Monash Data Futures Institute Seed Grants -  
AI and Data Science for Monash Global Challenges August 2021  
*Finding the right targets: most influential nodes in complex networks*  
Chief investigators: Ramialison, Currie, Dowe  
Adrian Salavaty is a named investigator in this grant.

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## PUBLICATIONS, PRESENTATIONS, AND ABSTRACTS

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### **Publications (Google Scholar Citations = 928)**

- **Salavaty A**, Douek AM, Kaslin J., Ramialison M, Currie PD. ExIR enables prioritizing driver and biomarker genes from omics data in a reference free manner. *iScience*. Jun 2026. PMID: [42291207](#).
- Rudraraju R., Gartner M. J., Neil J. A., Stout E. S., Chen J., Needham E. J., See M., Mackenzie-Kludas C., Lee L. Y. Y., Wang M., Pointer H., Karavendzas K., Abu-Bonsrah D., Drew D., Sun Y. B. Y., Tan J. P., Sun G., **Salavaty A**, *et al*. Parallel use of human stem cell lung and heart models provide insights for SARS-CoV-2 treatment. *Stem Cell Reports*. Jun 2023. PMID: [37315523](#)
- Ruparelia A, **Salavaty A**, *et al*. The African killifish: A short-lived vertebrate model to study the biology of sarcopenia and longevity. *Aging Cell*. May 2023. PMID: [37183563](#)
- **Salavaty A**, Esmaeel Azadian, Shalin H. Naik, Currie P. Clonal selection parallels between normal and cancer tissues. *Trends in Genetics*. February 2023. PMID: [36842901](#)

- **Salavaty A**, Sara Alaei Shehni, Ramialison M, Currie P. Systematic molecular profiling of acute leukemia cancer stem cells allows identification of druggable targets. *Heliyon*. October 2022. PMID: [36281397](#)
- **Salavaty A**, Ramialison M, Currie P. Integrated Value of Influence: An Integrative Method for the Identification of the Most Influential Nodes within Networks. *Patterns*. August 2020. PMID: [33205118](#)
- **Salavaty A**, Rezvani Z, Najafi A. Survival analysis and functional annotation of long non-coding RNAs in lung adenocarcinoma. *Journal of Cellular and Molecular Medicine*. June 2019. PMID: [31211495](#)
- **Salavaty A**, Movahedi Motlagh F, *et al.* Potential role of RAB6C-AS1 long noncoding RNA in different cancers. *Journal of Cellular Physiology*. August 2018. PMID: [30076712](#)
- **Salavaty A**, Mohammadi N, Shahmoradi M, Naderi Soorki M. Bioinformatic analysis of circadian expression of oncogenes and tumor suppressor genes. *Bioinformatics and Biology Insights*. December 2017. PMID: [29276378](#)
- Hajjari M, Sadeghi I, **Salavaty A**, Nasiri H, Birgani MT. Tissue Specific Expression Levels of Apoptosis Involved Genes Have Correlations with Codon and Amino Acid Usage. *Genomics & Informatics*. 14(4):234-240. December 2016. PMID: [28154517](#)
- Hajjari M, **Salavaty A**, Crea F, Shin YK. The potential role of PHF6 as an oncogene: a genotranscriptomic/proteomic meta-analysis. *Tumor Biology*. 37:5317-5325. April 2016. PMID: [26561469](#)
- **Salavaty A**. Carcinogenic effects of circadian disruption: an epigenetic viewpoint. *Chinese Journal of Cancer*. 34-38. June 2015. PMID: [26253128](#)
- Hajjari M, **Salavaty A**. *HOTAIR*: an oncogenic long non-coding RNA in different cancers. *Cancer Biology & Medicine*. 12:1-9. March 2015. PMID: [25859406](#)

### **Working Manuscripts**

- **Salavaty A**, Foroutan M, Sandhu S, Molania R. ClustoCell reveals cell types and states from single-cell transcriptomes.

### **Presentations and Abstracts**

- International Congress of Genetics (ICG) July 2023  
Melbourne Convention and Exhibition Centre (MCEC), Melbourne, Australia  
- Lightning Talk and poster presentation: **Salavaty A**, Pinese M. InCRIMP: a versatile computational model for the integrative analysis of multi-omics data.
- Australasian Genomic Technologies Association (AGTA) October/November 2022  
Sunshine Coast, Queensland, Australia  
- Poster presentation: **Salavaty A**, Pinese M. InCRIMP: a versatile computational model for the integrative analysis of multi-omics data.

- Oral presentation: **Salavaty A**, Ramialison M, Currie P. Identification, classification, and prioritization of most influential players in normal biological processes and diseases.
  - COMBINE-ABACBS Student Symposium November 2021
  - Victorian Cancer Bioinformatics Symposium October 2021
  - Oz Single Cell – PERTH September 2021
  - ANZSCDB August 2021
- ISMB/ECCB conference alliance July 2021  
Virtual
  - E-Poster and Short Talk: **Salavaty A**, Ramialison M, Currie P. Identification, classification, and prioritization of most influential players in normal biological processes and diseases.
  - E-Poster and Short Talk: **Salavaty A**, Ramialison M, Currie P. Identification of the most influential nodes involving all topological dimensions of a network.
- 5<sup>th</sup> International Conference on Proteomics & Bioinformatics September 2015  
OMICS Group, Valencia, Spain  
Abstract: **Salavaty A**, Hajjari M. Meta-analysis of RAG2 using a genotranscriptomic/proteomic approach: suggestive of its oncogenic role. *Journal of Proteomics & Bioinformatics*. 8:88. October 2015. DOI: 10.4172/0974-276X.S1.077

### **Books**

- Co-translator of “Next Generation Sequencing Technologies in Medical Genetics” book  
ISBN: 978-600-356-502-9; National Bibliography Number (NBN): 4282468  
Shahid Chamran University of Ahvaz 2015  
Supervisor: Dr. Maryam Naderi Soorki

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### **RESEARCH INTERESTS**

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- Bioinformatics and systems biology analysis of cancer initiation/progression causes
- Cancer stem cells
- Epigenetic basis of carcinogenesis
- AI for Biomedicine
- Development of tools and packages for the identification, classification, prioritization, and visualization of biological molecules

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### **APPENDICES**

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#### **Awards and Honors**

- Ph.D. scholarship from the Vancouver Prostate Center, University of British Columbia, (offered, but not taken up) 2018
- Ph.D. scholarship from Griffith University, (offered, but not taken up) 2018

- Ph.D. scholarship from the John Curtin School of Medical Research, Australian National University, (offered, but not taken up) 2018
  - Ranked 2<sup>nd</sup> among all M.Sc. students in Biochemistry, University of Kashan 2018
  - Ranked among the top 4% of participants in the National University Entrance Exam, Iran 2011
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## **Publications, Presentations, and Abstracts**

### **Publications**

- **Salavaty A**, Rezvani Z, Najafi A. Long non-coding RNA LINC00987 may function as a tumor suppressor in lung adenocarcinoma. *F1000Research*. May 2018. DOI: [10.12688/f1000research.14785.1](https://doi.org/10.12688/f1000research.14785.1)
- Sedghi M, Esfandiari E, Fazel-Najafabadi E, Salehi M, **Salavaty A**, Fattahpour S, Dehghani L, Nouri N, Mokarian F. Genomic rearrangement screening of the BRCA1 from seventy Iranian high-risk breast cancer families. *Journal of Research in Medical Sciences*. 21:95. November 2016. PMID: [28163741](https://pubmed.ncbi.nlm.nih.gov/28163741/)

### **Presentations**

- Global Meet on Nanomedicine & Healthcare November 2017  
New Orleans, USA  
E-Poster: **Salavaty A**, Shahmoradi M. Application of nanobioinformatics in drug design and delivery systems. *Biol Med Case Rep*. DOI: [10.13140/RG.2.2.24092.39043](https://doi.org/10.13140/RG.2.2.24092.39043)
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## **Professional Experiences**

### **Affiliations**

- Member of Complex Biological Systems Alliance (CBSA), 2015-present  
a global non-profit research consortium

### **Teaching Experiences**

- Manuscript Writing Workshop 2016  
University of Kashan Kashan, Iran
- Secondary school-level courses in English language 2015  
Ghلامchi Educational Foundation Isfahan, Iran